Area-efficient and Low-power Face-to-Face-bonded 3D Liquid State Machine Design

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ABSTRACT

As small-form-factor and low-power end devices matter in the cloud networking and Internet-of-Things Era, the bio-inspired neuromorphic architectures attract great attention recently in the hope of reaching the energy-efficiency of brain functions. Out of promising solutions, a liquid state machine (LSM), that consists of randomly and recurrently connected reservoir neurons and trainable readout neurons, has shown a great promise in delivering brain-inspired computing power. In this work, we adopt the state-of-the-art face-to-face (F2F)-bonded 3D IC flow named Compact-2D [4] to the LSM processor design, and study the power-area-accuracy benefits of 3D LSM ICs targeting the next generation commercial-grade neuromorphic computing platforms. First, we analyze how the different size and connection density of a reservoir in the LSM architecture affects the learning performance using the real-world speech recognition benchmark. Also, we explore how much the power-area design overhead should be paid off to enable better classification accuracy. Based on the power-area-accuracy trade-off, we implement a F2F-bonded 3D LSM IC using the optimal LSM architecture, and finally justify that 3D integration practically benefits the LSM processor design in huge form factor and power savings while preserving the best learning performance.

CCS CONCEPTS

• **Computer systems organization → Neural networks**; • **Hardware → Neural systems**;

KEYWORDS

3D IC; Face-to-Face Bonding; Liquid State Machine;

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1 INTRODUCTION

Today's Internet-of-Things paradigm boosted by the 5th generation wireless technology handles formidable data provided from us, and demands more efficient visual and auditory information processing at the edge computing level. As small form factor and low power consumption is found critical, significant research efforts have been made on spiking neural networks (SNNs) due to their biological plausibility and energy efficiency. Among noticeable approaches using SNNs, the liquid state machine (LSM) has been featured as a special competent for spatiotemporal pattern classification, such as speech recognition and bio-signal processing thanks to the architectural simplicity and computational efficiency [2, 6, 12, 13].

In this work, we adopt face-to-face(F2F)-bonded 3D integration technology to the LSM neuromorphic processor design. The advancement of F2F wafer-level bonding technology has improved the bonding precision significantly ($\leq 0.5\mu m$ pitch [1]), and this allows us to enable fine-grained 3D interconnections to maximize 3D IC benefits. Using the state-of-the-art RTL-to-GDSII physical design flow named Compact-2D [4], we explore the power-area-accuracy benefits of F2F-bonded 3D LSM processors targeting the next generation neuromorphic processors.

2 LIQUID STATE MACHINE

2.1 System Architecture

Structurally speaking, the LSM architecture consists of a reservoir and a readout stage. In the reservoir, neurons are randomly and recurrently connected like a liquid pond, or a reservoir. When the input patterns are fed into the reservoir stage, interactive spike signaling between recurrently connected reservoir neurons transfers the input patterns to a high-dimensional non-linear space. These preprocessed input patterns are used for the supervised spike-timing-dependent plasticity (STDP) learning in the readout stage for the final classification.

The operations of neurons are executed in parallel synchronized by the global finite state machine. Each digital neuron contains three functional modules: a synaptic input processing module that generates a synaptic response for an input spike; a spike generation module that accumulates the synaptic responses from all presynaptic neurons and updates the membrane potential based on the leaky integrate-and-ïňĄre model; and a learning module that tunes the afferent presynaptic weights of the associated neuron. Thanks to the activity-dependent clock gating [6], the clock signal of each functional module only toggles when the module needs to be activated. Also, we adopt the memory sharing mechanism [5] to further reduce the power and area overhead of memory macros.

2.2 Training Algorithms

STDP is an unsupervised Hebbian learning mechanism updating synaptic weights based on the temporal relationship of pre- and postsynaptic spikes:

$$\begin{aligned} \Delta w^{+} &= A_{+}(w) \cdot e^{-\frac{|\Delta t|}{\tau_{+}}} \ \ if\, \Delta t > 0 \\ \Delta w^{-} &= A_{-}(w) \cdot e^{-\frac{|\Delta t|}{\tau_{-}}} \ \ if\, \Delta t < 0, \end{aligned} \tag{1}$$

where Δw^{+} and Δw^{-} are the weight modifications induced by long-term potentiation (LTP) and long-term depression (LTD), and $A_{\pm}(w)$ determines the strength of LTP/LTD.

Accurate STDP implementation produces good performance, however, at a cost of highly discretized bit resolution and frequent weight updates, resulting in the large area/power overhead. However, the aggressively discretized STDP learning curve leads to immediate performance degradation. To address this problem, we adopt look-up table based STDP implementation with minimal aggregated discretization error [3] for the reservoir training until the synaptic weight distribution converges.

During the readout training phase, reservoir stage continues to be activated to provide spike inputs to training stage while maintaining its synaptic weights. Then a biologically plausible supervised spike-based algorithm [13] is employed at the readout stage to perform the classification. To introduce the supervision on spikes, an additional stimulus is injected into each readout neuron to bring up the firing activity to the desired level, which is characterized biologically by a Calcium concentration model. The Calcium concentration C is computed by:

$$C(t) = C(t-1) - \frac{C(t-1)}{\tau_c} + S(t), \tag{2}$$

where $S(t)$ is the spiking event at the current time step. Then, the readout synapse weight is update statistically:

$$\begin{aligned} w_i &= w_i + \Delta w \ with\, P_{+},\ if\, C_{\theta} < C < C_{\theta} + \Delta C \\ w_i &= w_i - \Delta w \ with\, P_{-},\ if\, C_{\theta} - \Delta C < C < C_{\theta}, \end{aligned} \tag{3}$$

where P_{+} and P_{-} are the probabilities for potentiation and depression, and C_{θ} and ΔC are the calcium concentration threshold and margin width respectively.

3 PHYSICAL DESIGN METHODOLOGY

In this work, we adopt the state-of-the-art F2F-bonded 3D IC design flow named Compact-2D [4] and design power-area efficient two-tier F2F-bonded 3D LSM ICs. Compact-2D is a full-chip RTL-to-GDSII physical design solution that builds high-quality F2F-bonded 3D ICs utilizing the commercial 2D place and route (P&R) engines. Each design step of Compact-2D suggests a unique method to manipulate 2D P&R engines for the optimal 3D placement and routing solution.

3.1 3D Placement Using 2D IC Engines

For the 3D placement, Compact-2D flow proposes the two design steps: Compact-2D design and tier partitioning. Compact-2D design decides the X, Y 2D locations of standard cells using conventional 2D P&R steps based on the specially treated floorplan, then the following tier partitioning decides the Z location of cells. It is worth noting that the target F2F floorplan determined in the 3D floorplan stage is too small to accommodate all the synthesized gates of the design. Therefore, Compact-2D design requires to expand the target F2F floorplan up to the size meeting the initial placement utilization of 2D IC counterpart. However, the wirelength of a net on the Compact-2D design is also longer than that of the target F2F design due to the expanded floorplan, and this requires to scale down the interconnect RC parasitics by a factor of the floorplan expansion. Interconnect RC scaling allows to imitate the net parasitics of final F2F design, and removes or downsizes redundant buffers for the timing closure.

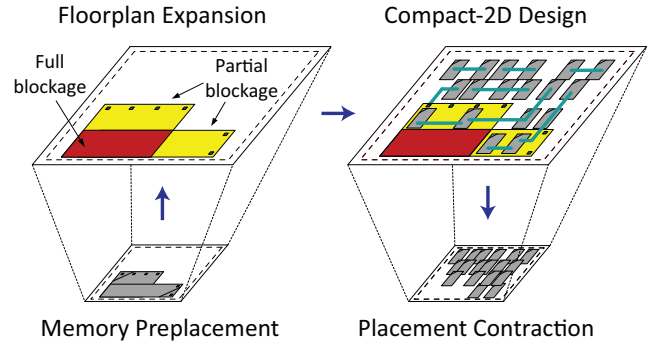

Figure 1: Design steps in the Compact-2D flow [4]. Compact-2D design utilizes an expanded F2F design floorplan and scaled interconnect RC parasitics (in green). Then, the placement contraction step finalizes the X, Y location, and subsequent tier-partitioning determines the Z location of 3D placement.

In the Compact-2D design, the size of memory blocks turns into the minimum placement unit, and the original memory area is covered by placement blockages. For the fully overlapped region, full placement blockage is used to prevent from the standard cell placement, and for the partially overlapped regions, partial blockages (40% in this work) are used to enable the cell placement. The cells in the partial blockage regions are naturally located to the tier where the memory does not occupy in a two-tier F2F design. After the Compact-2D design is finished, the X,Y locations of standard cells is mapped into the target F2F design footprint to finalize the X,Y solution of 3D placement. This is called placement contraction. Figure 1 demonstrates the overall steps from floorplan expansion to the placement contraction. To decide the Z location of cells, bin-based Fiduccia-Mettheyses (FM)-mincut tier partitioning [9] is followed based on the placement contraction result.

3.2 3D Routing Using 2D IC Engines

For the 3D routing, 3D technology Library Exchange Format(LEF) and macro LEF should be prepared. 3D LEF contains all the metal stack for the 3D routing including F2F via definition and design rules, and 3D macro LEF instantiates the cells based on their tier location. Therefore, all though they are on the same placement area, the metal layer of the cell pins on the different tiers is physically separated. Compact-2D supports post-tier-partitioning optimization that optimizes the buffers to meet the timing in the 3D routing environment. Post-tier-partitioning optimization requires the legal

placement solution on the small target F2F design footprint. This is done by placement row splitting, which means the height of a cell and a placement row is halved assuming two tiers. The pin locations of each cell is, however, preserved outside the cell region for the correct 2D timing engine performance.

Once the post-tier-partitioning optimization is done, 3D nets are sliced into the small subnets for each tier depending on their running metal layer. In this step, final F2F via location is fixed, and separate DEF and verilog files for each die are generated while the F2F vias are treated as the primary input/output ports. Then, incremental routing step recycles the routing result from post-tier-partitioning for the final tier-by-tier routing, and generates the signoff GDSII file for each die.

4 DESIGN AND SIMULATION SETTING

4.1 LSM Design Generation

Four LSM designs with different reservoir sizes are generated to study the impact of reservoir size. Each external input spike is randomly connected to 8 reservoir neurons with a fixed weight, and this fixed weight is randomly chosen to be 8 or -8 with equal probability. For internal connections in the reservoir stage, we follow the widely used settings suggested in [8], which is based on the microcircuit in the real biological brain in that 80% of the reservoir neurons are excitatory and 20% of the reservoir neurons are inhibitory. Synaptic connections are initialized stochastically according to the Euclidean distance between pre- and post-synaptic neurons. The probability of creating a connection between the neuron u and v is calculated by:

$$p = C \cdot exp\left[-(\frac{D(u,v)}{\lambda})^2\right]. \tag{4}$$

where λ is a connection parameter, and $D(u, v)$ is the Euclidean distance between the neuron u and v.

To study how the connectivity density affects the performance and the hardware costs of the LSM processor, we set the two representative values of parameter C for each network size, which are 1.5 and 4.5. As a result, the design with $C = 4.5$ has roughly three times more reservoir synapses than the design with $C = 1.5$. It is worth noting that the number of registers that stores the variables corresponding to the synapses is the same among all reservoir neurons and decided by the maximum reservoir synapse connections. However, as introduced in Equation 4, the synaptic connection between two neurons is a random variable, it is possible that a network actually instantiates more resources to implement the synapses although it has a smaller reservoir size. Table 1 tabulates the list of key components for the reservoir neuron implementation from four different benchmarks.

4.2 LSM Performance Simulation Setting

To measure the learning performance of proposed neural processors, we choose a representative non-trivial real-world benchmark of speech recognition, which is a subset of the TI46 speech corpus [11]. The benchmark has 260 speech samples of ten spoken utterances of English letters from "A" to "Z". The continuous temporal speech signals are preprocessed by Lyon's ear model [7] and then encoded into 78 spike trains using the Bens Spiker Algorithm [10]. We adopt a 5-fold cross validation scheme in the standard machine learning process to assess the learning performance.

Table 1: Reservoir neuron implementation. RVN72 denotes a design with 72 reservoir neurons in the reservoir stage, etc.

LSM Designs	RVN72	RVN90	RVN112	RVN135
Basedline Design: $C = 1.5$ **in Equation (4)**				
Shift Register	22	18	20	16
Input Synapse Width	4	4	5	5
Reservoir Synapse Width	5	5	5	4
Input Spike	16	16	21	32
Reservoir Spike	22	18	20	16
Dense Reservoir Design: $C = 4.5$ **in Equation (4)**				
Shift Register	31	38	48	35
Input Synapse Width	4	4	5	5
Reservoir Synapse Width	5	5	6	6
Input Spike	16	16	21	32
Reservoir Spike	31	38	48	35

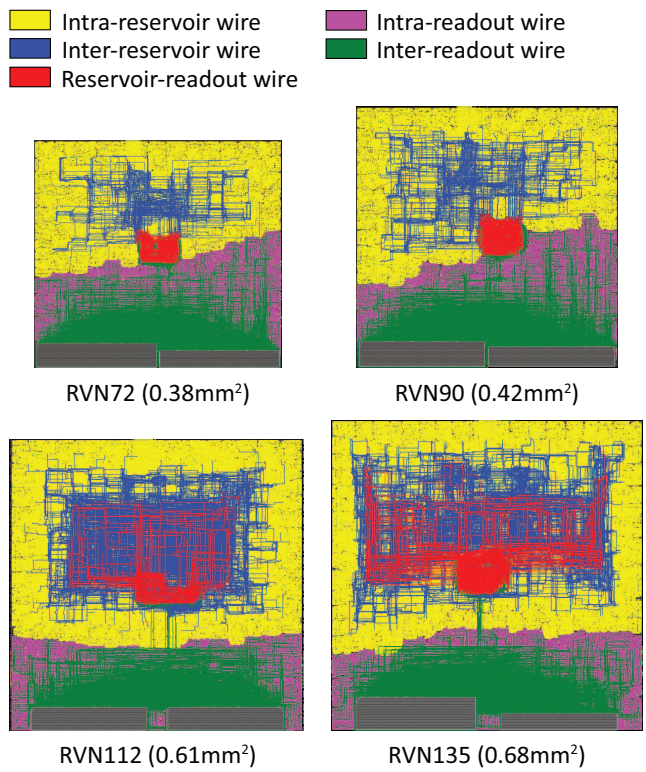

Figure 2: 2D full-chip LSM designs.

5 2D IC DESIGN RESULTS

5.1 Impact of Reservoir Size

For the full-chip 2D LSM designs, we use four metal layers in a commercial-grade 28nm Process-Design-Kit, and use different sizes of register file for the shared memory in the readout stage depending on the reservoir size. The maximum target clock frequency is

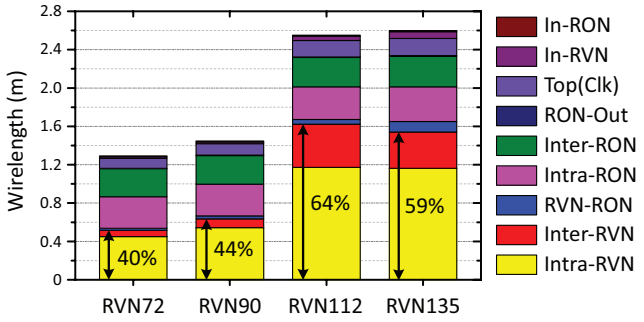

Figure 3: Wirelength distribution of the 2D LSM designs. RVN and RON denotes reservoir and readout neurons, respectively. As the reservoir size increases, wirelength from the reservoir network becomes dominant.

1.3GHz for all the designs. Figure 2 shows the silicon area and the interconnections colored by the characteristics of connectivity in four different 2D LSM designs. The number of reservoir neurons in the reservoir stage is used to name those designs. The RVN135 LSM design requires 78% more silicon area, and 101% more total wirelength compared with those of RVN72 for the 2D full-chip implementation.

Figure 3 shows the detailed wirelength distribution divided by the purpose of the interconnection. As the reservoir size increases, not only the wirelength for the intra-RVN is increased in proportion to the reservoir population, but also the inter-RVN wirelength is increased since the reservoir neurons are spread out of the entire design. This contributes to additional wirelength for full-chip implementation. Compared to RVN72, the inter-RVN wirelength for the RVN135 design is exponentially increased by 5.7x while intra-RVN wirelength is increased by 2.6x. In addition, it is worth noting that RVN112 has only 1.8% less total wirelength while the silicon area is 10.3% smaller than RVN135. This is because RVN112 has inherently larger reservoir spike connectivity than the other designs as presented in Table 1. It turns out that RVN112 actually has 19.3% more wirelength for inter-reservoir neuron connections than RVN135 design. This indicates that the fine-tuned parameter optimization for the initial network generation is critical for the area-efficient low-power LSM designs targeting the edge-computing devices.

To summarize the impact of reservoir size on the power-area-accuracy of 2D LSM designs, Table 2 tabulates the detailed design and performance metrics. The target clock frequency is 1.3GHz, and the static power analysis is performed based on the 0.1 switching activity for the primary inputs and D-Flip Flop outputs, and 2.0 for the clock. The placement utilization of all designs is targeted within 75% 78% range for the fair design comparison. We first observe that larger reservoir network size gives us the better classification accuracy. RVN135 design achieves 92% speech recognition accuracy while RVN72 design has only 88% classification performance. This makes sense since a reservoir with more neurons holds a more complex non-linear dynamics, and can map the input patterns to a higher-dimensional feature space which provide rich information for the readout layer for classification. However, this improvement

Table 2: 2D LSM designs with coarser reservoir connectivity. The target clock frequency is 1.3GHz. RVN135 design shows 1.78x more silicon area, and 1.96x more power consumptions compared with RVN72 design, while improving the classification accuracy.

Baseline Neuron Connectivity				
2D LSM Designs	RVN72	RVN90	RVN112	RVN135
Reservoir Neuron Count	72	90	112	135
Worst Neg. Slack (ps)	1.28	1.60	13.01	12.72
Classification Accuracy	87.69%	88.85%	90.76%	91.54%
Placement Util. (%)	75.10	76.50	78.02	77.63
Silicon Area (mm^2)	0.378	0.421	0.613	0.678
Total Wirelength (m)	1.292	1.447	2.552	2.598
Seq. Cell Count	6,893	7,966	14,406	15,572
Comb. Cell Count	81,043	91,284	135,871	150,517
Total Cell Count	87,936	99,250	150,277	166,089
Switching Pwr (mW)	46.88	54.03	86.62	90.92
Internal Pwr (mW)	21.43	24.86	39.38	42.74
Leakage Pwr (mW)	6.25	7.26	11.64	12.75
Total Pwr (mW)	74.56	86.15	137.64	146.41

is not for free with regard to the silicon area, and total power consumption. Compared with RVN72, 78% of form factor increase in RVN135 leads to more timing buffers on top of the increased wirelength, and they contribute to 96.4% more total power consumption in RVN135.

5.2 Impact of Reservoir Connectivity

We observe that the increased reservoir connectivity affects the intra-reservoir neuron wirelength. This is because each reservoir neuron contains more shift registers and reservoir spike storages in its implementation to support the dense reservoir connecitivity as presented in the Table 1. The intra-reservoir neuron wirelength is increased by 67% in average. Regarding the inter-reservoir neuron wirelength, we observe 3.69x more wirelength in the dense reservoir designs in average. A maximum 5.42x increase is observed in RVN90 design, and a minimum 1.84x increase in RVN112 due to RVN112's inherent enhanced reservoir connectivity density decided by the random process when we generate the design.

To analyze the power-area-accuracy impact of the reservoir connectivity density, Table 3 tabulates the performance and design metrics of dense reservoir designs. For the fair comparison, the metal layer usage, maximum target clock frequency, and the static power analysis setting is the same with the baseline designs. First, we observe increased reservoir connectivity density gives us additional classification accuracy improvement without increasing the reservoir size. when the recurrent reservoir connection gets denser, the diversity of reservoir dynamics is improved with a better interaction among input samples, thus boost the learning performance. However, if the recurrent connections in a reservoir is too dense, there would be a chaos inside the reservoir and no performance improvement will be observed. Therefore, we do not see an obvious performance improvement for larger network size in which the overall synaptic connections are already large. Although this small

Table 3: 2D LSM designs with denser reservoir connectivity. Δ denotes an increase compared with the baseline connectivity. We observe slight increase in accuracy at the expense of 1.28x more silicon area and 1.38x more power consumption.

Dense Neuron Connectivity				
2D LSM Designs	RVN72	RVN90	RVN112	RVN135
Worst Neg. Slack (ps)	-11.98	-9.94	-14.65	-11.77
Classification Accuracy	88.46%	89.23%	90.77%	91.54%
Placement Util. (%)	77.86	77.65	78.39	78.20
Silicon Area (mm^2)	0.492	0.596	0.757	0.814
Total Wirelength (m)	2.019	2.490	3.329	3.526
Seq. Cell Count #	11,709	15,292	20,679	20,919
Comb. Cell Count #	110,424	129,489	164,456	180,230
Total Cell Count #	122,133	144,781	185,135	201,149
Switching Pwr (mW)	66.51	83.00	108.87	116.69
Internal Pwr (mW)	30.67	39.04	51.21	54.16
Leakage Pwr (mW)	9.17	11.43	15.30	16.22
Total Pwr (mW)	106.36	133.47	175.38	187.07
Total Pwr Δ	1.43x	1.55x	1.27x	1.28x

accuracy improvement is significant and impressive in the speech recognition field and also for the edge-computing devices, it turns out the area-power expense to enable the dense reservoir connectivity is not ignorable. We find that 1.28x more silicon area and 1.38x more power consumptions is required in average for the designs with denser reservoir connectivity compared with their baseline. This indicates that if we put the accuracy to the highest priority on the LSM design, larger reservoir network with reasonable reservoir connectivity offers better power-area-accuracy tradeoffs in LSM design implementation.

6 3D IC DESIGN RESULTS

To preserve the accuracy benefit while minimizing the form factor and power consumption, we build two-tier F2F-bonded 3D LSM designs. For the experiments, we focus on the RVN135 architecture with baseline reservoir connectivity, which gives the maximum classification performance among the different reservoir sizes, and better area-power results compared with the dense reservoir connectivity. For the full-chip two-tier F2F-bonded 3D LSM designs, four metal layers are used for each die. F2F via size is assumed to be 0.5um, the pitch is 1.0um, resistance is 0.2ohm and 0.5fF for via capacitance. For the initial floorplan, two register files are split into different tiers and vertically overlapped to minimize the form factor occupied by them.

In Table 4, we present 2D vs. F2F-bonded 3D RVN135 designs at iso-performance. For fair comparison, placement utilization for 2D and both dies of F2F design is set to be the same. First, we achieve more than 50% footprint saving with F2F compared with 2D. This leads to 4% silicon area saving under comparable placement density (78.63% vs. 79.06%). We believe this is significant as most of the work published in the literature show 50% footprint saving, which means zero silicon area saving. Moreover, we achieved this area saving under the identical classification accuracy. We believe such a footpring saving is useful for Internet-of-Things sensors that

Table 4: F2F-bonded 3D RVN135 vs. 2D RVN135. F2F achieves 52% form factor savings, 4% silicon area savings, and 3% total power savings under the same 92% accuracy. Δ% denotes savings of the F2F-bonded 3D IC.

Designs	2D	F2F	Δ%
Worse Neg. Slack (ps)	-12.72	-9.53	-
Form Factor (mm^2)	0.678	0.326	51.9%
Silicon Area (mm^2)	0.678	0.652	3.8%
Placement Util. (%)	78.63	79.06	-0.5%
Total Wirelength (m)	2.598	2.119	18.4%
Seq. Cell Count	15,572		0.0%
Comb. Cell Count	150,517	144,922	3.7%
Total Cell Count	166,089	160,494	3.4%
Pin Capacitance (pF)	436.7	423.5	3.0%
Wire Capacitance (pF)	232.6	199.2	14.4%
Total Capacitance (pF)	669.3	622.7	7.0%
Wire Cap Ratio (%)	34.95	31.99	8.5%
Switching Power (mW)	90.92	86.45	4.9%
Internal Power (mW)	42.74	42.34	0.9%
Leakage Power (mW)	12.75	12.91	-1.3%
Total Power (mW)	146.41	141.90	3.1%
Accuracy (%)	92%		-

require smallest possible form factor. The silicon area saving directly impacts cost. The wirelegnth (2.598 vs. 2.119) and cell count saving (166K vs. 160K) in F2F leads to 3% total power saving. Power saving is not as significant as our LSM architecture is pin-cap dominated: in 2D design, pin:wire capacitance ratio is 436.7: 232.6. In 3D design, 423.5: 199.2. The wirelegnth savign in 3D design only affects wire capacitance, thus the small power saving.

Another reason for small power saving is due to the cell displacement introduced during Compact-2D flow. In Compact-2D, cell overlap after tier partitioning is removed using a legalizer, which necessitates cell displacement in both tiers. Figure 4 shows the difference between the X, Y cell location after tier partitioning and those of the final F2F design. while 34.8% cells keep the optimal placement location, 63.6% of cells change their location from $1\mu m$ to $4\mu m$. 0.19% of cells more than $10\mu m$ displacement while the post-tier-partitioning optimization. The total displacement is $0.21m$, and this partially accounts the wirelength increase (and power increase) in the final F2F design.

Figure 5 presents wirelength distribution comparisons. The wirelength of Compact-2D nets is scaled by 0.693 to show the difference between the electrical length of the Compact-2D nets and the actual wirelength of F2F nets. Most of small nets less than $20\mu m$ has negligible impact on the final F2F nets, but 1% of these small nets are affected by 3D routing and the resulting wirelength become larger than $20\mu m$. This is not considered during the Compact-2D design, and the nets between $20\mu m$ and $40\mu m$ has increased by 18%. Along with the placement difference, this 3D routing causes wirelength increase in F2F design, and looses the wire capacitance savings, and switching power savings obtained in the Compact-2D design. The final total power savings of F2F design turns into the 3%, with the significant form factor savings by 52%.

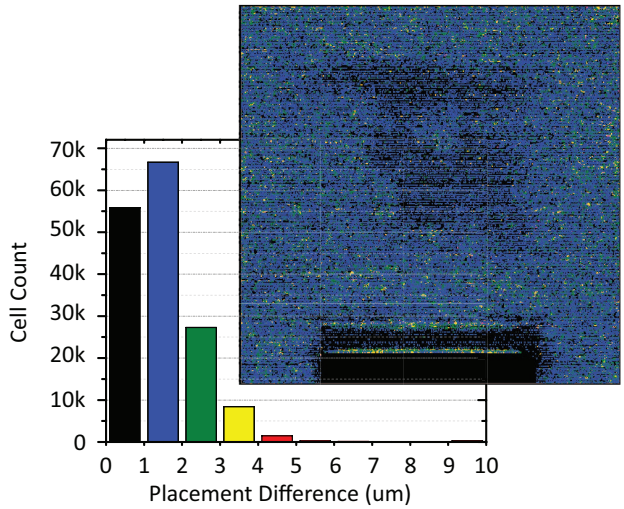

Figure 4: Cell displacements before and after tier partitioning. Cells are moved to remove overlaps caused by the placement contraction in Compact-2D [4]. We observe that 65.2% of the cells change their location. The total displacement is 0.21m. Green and yellow cells in the die shot have major displacements.

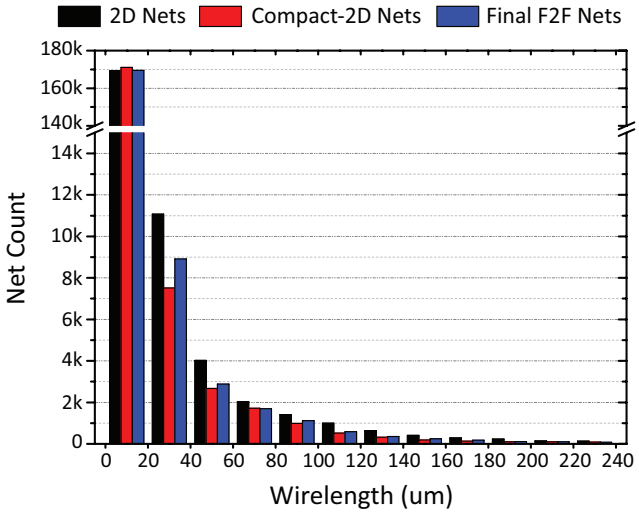

Figure 5: Wirelength distribution comparison among 2D, Compact-2D, and final F2F-bonded 3D LSM designs.

7 CONCLUSION

In this work, we studied the impact of the reservoir size and the connectivity density on the classification accuracy and their area-power overhead of the liquid state machine (LSM) processor designs. We showed that the 135 reservoir neurons, which is the maximum size of the reservoir used in this work, significantly improves the learning performance of a representative non-trivial real-world benchmark of speech recognition by 4.4% compared with the design with 72 reservoir neurons. However, we observed that this accuracy improvement is not for free and requires 78% more form factor, and 96.4% power consumption. Regarding the reservoir connectivity, we presented that denser reservoir improves the accuracy by a maximum 0.8% without increasing the network size, but this results in 28% more form factor, and 38% more power consumption in average.

Lastly, we explored the F2F-bonded 3D integration benefits on the LSM processor design using the state-of-the-art physical design flow name Compact-2D, and thoroughly analyzed the area-power benefits in the two-tier F2F-bonded 3D LSM processor design. We observed that F2F-bonded 3D integration has significant benefits on the form factor savings, and additional power savings while preserving the great classification accuracy. Using the design with 135 reservoir neurons, the F2F-bonded 3D LSM processor achieved 52% form factor savings, which is smaller than the 2D design with 72 liquid neurons by 13.8%, and additional 3% power savings compared to its 2D counterpart. This work suggests the power-area-accuracy tradeoffs on the reservoir optimization in the 2D and F2F-bonded 3D LSM processors targeting the next generation neuromorphic edge-computing devices in the Internet-of-Things Era.

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